A Data-Driven Assessment of Water Quality across Indian Aquatic Systems Using Multi-Parameter Environmental Indicators and Machine Learning

1. Abstract

Water quality assessment is essential for environmental sustainability and public health, particularly in India where diverse water bodies face increasing contamination from industrial discharge, agricultural runoff, and untreated sewage. This project presents a data-driven analysis of Indian water quality using machine learning and statistical techniques applied to multi-source environmental datasets.

The study involves data preprocessing, exploratory data analysis, Water Quality Index (WQI)-based assessment, unsupervised clustering using K-Means to identify pollution patterns, and anomaly detection using Isolation Forest to detect unusual contamination behavior across states and water bodies. An exploratory Random Forest classification model was also evaluated to assess predictive separability within the dataset.

The analysis reveals significant regional variation in pollution indicators such as dissolved oxygen, biological oxygen demand, nitrate concentration, and fecal coliform levels. Cluster-based analysis highlights distinct water quality profiles, while anomaly detection identifies potentially critical pollution events requiring further investigation.

The findings demonstrate how machine learning can support environmental monitoring and provide scalable analytical insights for water resource management and policy planning in India.

2. Study hypothesis:

Water quality is a critical aspect for both drinking water and marine life, as various parameters significantly impact the suitability of water for these purposes. Here are the key parameters and their typical acceptable ranges or conditions for both drinking water and marine life:

Drinking Water Quality Parameters

1. **Temperature**:
    - Generally, should be ambient and consistent. High temperatures can affect the taste and increase the growth of microorganisms.

2. **Dissolved Oxygen (DO)**:
    - Typically, drinking water should have a DO level of at least 5-6 mg/L. Higher levels are better as they ensure water is fresh and taste is maintained.
3. **Potential of Hydrogen (pH)**:
    - The pH of drinking water should be between 6.5 and 8.5. This range ensures that the water is neither too acidic nor too alkaline.
4. **Conductivity Level**:
    - Should be below 400 µS/cm. High conductivity indicates high levels of dissolved salts, which can affect taste and health.
5. **Biochemical Oxygen Demand (BOD)**:
    - Should be less than 2 mg/L. Low BOD indicates low levels of organic matter and good water quality.
6. **Nitrate (NO3-N)**:
    - Should be less than 10 mg/L to avoid health issues, particularly for infants (risk of methemoglobinemia or "blue baby syndrome").
7. **Nitrite (NO2-N)**:
    - Should be less than 1 mg/L. Higher levels can cause health problems similar to nitrates.
8. **Faecal Coliforms**:
    - Should be absent in drinking water. Their presence indicates contamination by faecal matter and potential presence of pathogens.
9. **Total Coliforms**:
    - Should be less than 1 CFU/100 mL. Total coliforms are used as an indicator of overall water sanitation quality.

Marine Life Water Quality Parameters

1. **Temperature**:
    - Varies depending on the species and habitat. Generally, marine life thrives in water that is within the natural temperature range of their environment.
2. **Dissolved Oxygen (DO)**:
    - Should be at least 5 mg/L. Lower levels can cause stress or mortality in marine organisms.
3. **Potential of Hydrogen (pH)**:
    - Should be between 7.5 and 8.5. Marine organisms are sensitive to pH changes, which can affect their metabolism and reproduction.
4. **Conductivity Level**:
    - Varies widely depending on the marine environment. Generally, marine water conductivity is higher due to the presence of salts (typically around 50,000 µS/cm).

5. **Biochemical Oxygen Demand (BOD)**:
    - Should be less than 5 mg/L. Higher BOD can deplete oxygen levels, affecting marine life.
6. **Nitrate (NO3-N)**:
    - Should be less than 0.1 mg/L in open marine waters. Higher levels can lead to eutrophication, affecting the marine ecosystem.
7. **Nitrite (NO2-N)**:
    - Should be less than 0.02 mg/L. Nitrite is toxic to marine life, even at low concentrations.
8. **Faecal Coliforms**:
    - Ideally, should be absent, especially in areas where marine life is harvested for human consumption. Presence indicates contamination and potential health risks.
9. **Total Coliforms**:
    - Should be minimal, particularly in sensitive or protected marine areas. High levels indicate pollution and can affect the health of marine ecosystems.

Summary

- **Drinking Water**: Focuses on safety and health standards, with strict limits on contaminants and pollutants.
- **Marine Life**: Emphasizes maintaining natural conditions and avoiding pollutants that can disrupt marine ecosystems.

Ensuring water quality for both drinking and marine life involves regular monitoring and adherence to established guidelines and standards set by health and environmental authorities.

3. Literature Review:

Water Quality in India: State-wise and River-wise Analysis

Andhra Pradesh

- **Rivers:** Godavari, Krishna, Penna
- **Water Quality Issues:** High levels of pollutants due to industrial effluents, untreated sewage, and agricultural runoff.
- **Key Pollutants:** Biological oxygen demand (BOD), nitrates, phosphates, heavy metals.

Arunachal Pradesh

- **Rivers:** Siang, Subansiri, Lohit
- **Water Quality Issues:** Relatively unpolluted due to low industrial activities but affected by occasional sedimentation from natural erosion.
- **Key Pollutants:** Sediments.

Assam

- **Rivers:** Brahmaputra, Barak
- **Water Quality Issues:** Pollution from urban sewage, industrial discharge, and agricultural runoff.
- **Key Pollutants:** Heavy metals, pesticides, coliform bacteria.

Bihar

- **Rivers:** Ganga, Gandak, Kosi
- **Water Quality Issues:** High levels of faecal coliforms due to untreated sewage, agricultural runoff, and industrial discharge.
- **Key Pollutants:** BOD, coliform bacteria, nitrates.

Chhattisgarh

- **Rivers:** Mahanadi, Indravati, Hasdeo
- **Water Quality Issues:** Industrial discharge, mining runoff.
- **Key Pollutants:** Heavy metals, BOD, chemicals from mining activities.

Goa

- **Rivers:** Mandovi, Zuari
- **Water Quality Issues:** Pollution from mining activities, untreated sewage.
- **Key Pollutants:** Heavy metals, BOD, coliform bacteria.

Gujarat

- **Rivers:** Sabarmati, Tapi, Narmada
- **Water Quality Issues:** High industrial discharge, urban sewage.
- **Key Pollutants:** Toxic chemicals, heavy metals, BOD.

Haryana

- **Rivers:** Yamuna, Ghaggar
- **Water Quality Issues:** Industrial discharge, urban sewage.
- **Key Pollutants:** Heavy metals, pesticides, coliform bacteria.

Himachal Pradesh

- **Rivers:** Beas, Sutlej, Ravi
- **Water Quality Issues:** Pollution from hydropower projects, domestic sewage.
- **Key Pollutants:** Sediments, coliform bacteria.

Jharkhand

- **Rivers:** Damodar, Subarnarekha
- **Water Quality Issues:** Industrial discharge, mining runoff.
- **Key Pollutants:** Heavy metals, toxic chemicals.

Karnataka

- **Rivers:** Cauvery, Tungabhadra, Krishna
- **Water Quality Issues:** Pollution from industrial activities, urban sewage.
- **Key Pollutants:** BOD, heavy metals, nitrates.

Kerala

- **Rivers:** Periyar, Pamba, Bharathapuzha
- **Water Quality Issues:** Industrial effluents, untreated sewage.
- **Key Pollutants:** BOD, coliform bacteria, heavy metals.

Madhya Pradesh

- **Rivers:** Narmada, Chambal, Betwa
- **Water Quality Issues:** Industrial discharge, urban sewage.
- **Key Pollutants:** BOD, heavy metals, coliform bacteria.

Maharashtra

- **Rivers:** Godavari, Krishna, Ulhas, Mithi
- **Water Quality Issues:** Industrial discharge, urban sewage, agricultural runoff.
- **Key Pollutants:** Heavy metals, BOD, coliform bacteria.

Manipur

- **Rivers:** Imphal, Barak
- **Water Quality Issues:** Pollution from domestic sewage, agricultural runoff.
- **Key Pollutants:** BOD, coliform bacteria.

Meghalaya

- **Rivers:** Umngot, Myntdu
- **Water Quality Issues:** Mining runoff, domestic sewage.
- **Key Pollutants:** Heavy metals, coliform bacteria.

Mizoram

- **Rivers:** Tlawng, Tut
- **Water Quality Issues:** Relatively unpolluted but impacted by domestic sewage.
- **Key Pollutants:** Coliform bacteria.

Nagaland

- **Rivers:** Dhansiri, Doyang
- **Water Quality Issues:** Pollution from agricultural runoff, domestic sewage.
- **Key Pollutants:** Pesticides, coliform bacteria.

Odisha

- **Rivers:** Mahanadi, Brahmani, Baitarani
- **Water Quality Issues:** Industrial effluents, mining runoff.
- **Key Pollutants:** Heavy metals, BOD, nitrates.

Punjab

- **Rivers:** Sutlej, Beas
- **Water Quality Issues:** Industrial discharge, agricultural runoff.
- **Key Pollutants:** Pesticides, heavy metals, BOD.

Rajasthan

- **Rivers:** Chambal, Banas, Luni
- **Water Quality Issues:** Industrial discharge, agricultural runoff.
- **Key Pollutants:** Toxic chemicals, salinity, BOD.

Sikkim

- **Rivers:** Teesta, Rangit
- **Water Quality Issues:** Minimal pollution due to low industrial activity but affected by domestic sewage.
- **Key Pollutants:** Coliform bacteria.

Tamil Nadu

- **Rivers:** Cauvery, Cooum, Adyar
- **Water Quality Issues:** Industrial effluents, untreated sewage, agricultural runoff.
- **Key Pollutants:** BOD, heavy metals, coliform bacteria.

Telangana

- **Rivers:** Godavari, Krishna, Musi
- **Water Quality Issues:** Industrial effluents, untreated sewage.
- **Key Pollutants:** BOD, heavy metals, coliform bacteria.

Tripura

- **Rivers:** Gomati, Manu
- **Water Quality Issues:** Pollution from domestic sewage, agricultural runoff.
- **Key Pollutants:** BOD, coliform bacteria.

Uttar Pradesh

- **Rivers:** Ganga, Yamuna, Gomti
- **Water Quality Issues:** Industrial discharge, domestic sewage, agricultural runoff.
- **Key Pollutants:** BOD, faecal coliforms, heavy metals.

Uttarakhand

- **Rivers:** Ganga, Yamuna, Bhagirathi
- **Water Quality Issues:** Pollution from domestic sewage, tourism-related waste.
- **Key Pollutants:** Coliform bacteria, BOD.

West Bengal

- **Rivers:** Hooghly, Damodar, Subarnarekha
- **Water Quality Issues:** Industrial effluents, untreated sewage.
- **Key Pollutants:** Heavy metals, BOD, coliform bacteria.

Union Territories

- **Andaman and Nicobar Islands**
    - **Rivers:** None (mainly creeks and small streams)
    - **Water Quality Issues:** Minimal pollution but impacted by domestic waste.

    - **Key Pollutants:** Coliform bacteria.
- **Chandigarh**
    - **Rivers:** None (mainly small streams)
    - **Water Quality Issues:** Pollution from urban sewage.
    - **Key Pollutants:** Coliform bacteria, BOD.
- **Dadra and Nagar Haveli and Daman and Diu**
    - **Rivers:** Daman Ganga, Kolak
    - **Water Quality Issues:** Industrial effluents, urban sewage.
    - **Key Pollutants:** Heavy metals, BOD, coliform bacteria.
- **Lakshadweep**
    - **Rivers:** None (mainly groundwater and small lagoons)
    - **Water Quality Issues:** Minimal pollution due to low industrial activity.
    - **Key Pollutants:** Coliform bacteria.
- **Puducherry**
    - **Rivers:** None (mainly small streams and backwaters)
    - **Water Quality Issues:** Pollution from domestic sewage, agricultural runoff.
    - **Key Pollutants:** Coliform bacteria, BOD.

Major Rivers in India and Their Pollution Levels

Ganga

- **States:** Uttarakhand, Uttar Pradesh, Bihar, Jharkhand, West Bengal
- **Water Quality Issues:** High levels of BOD, faecal coliforms, industrial pollutants.
- **Key Pollutants:** Heavy metals, coliform bacteria, nitrates.

Yamuna

- **States:** Uttarakhand, Haryana, Delhi, Uttar Pradesh
- **Water Quality Issues:** Extremely high levels of faecal coliforms, industrial discharge, untreated sewage.
- **Key Pollutants:** BOD, coliform bacteria, heavy metals.

Godavari

- **States:** Maharashtra, Telangana, Andhra Pradesh
- **Water Quality Issues:** Industrial effluents, agricultural runoff.
- **Key Pollutants:** BOD, heavy metals, pesticides.

Krishna

- **States:** Maharashtra, Karnataka, Telangana, Andhra Pradesh
- **Water Quality Issues:** Industrial discharge, untreated sewage.
- **Key Pollutants:** BOD, heavy metals, coliform bacteria.

Narmada

- **States:** Madhya Pradesh, Gujarat
- **Water Quality Issues:** Industrial pollutants, agricultural runoff.
- **Key Pollutants:** Heavy metals, BOD, pesticides.

Brahmaputra

- **States:** Arunachal Pradesh, Assam
- **Water Quality Issues:** Relatively unpolluted but impacted by urban waste in some areas.
- **Key Pollutants:** Sediments, coliform bacteria.

Cauvery

- **States:** Karnataka, Tamil Nadu
- **Water Quality Issues:** Industrial discharge, urban sewage, agricultural runoff.
- **Key Pollutants:** BOD, heavy metals, coliform bacteria.

Mahanadi

- **States:** Chhattisgarh, Odisha
- **Water Quality Issues:** Industrial effluents, mining runoff.
- **Key Pollutants:** Heavy metals, BOD, nitrates.

Solutions and Efforts

National Mission for Clean Ganga (NMCG)

- **Objective:** Rejuvenation of the Ganga River through comprehensive measures.
- **Initiatives:** Establishing sewage treatment plants, river surface cleaning, rural sanitation.

Pollution Control Boards

- **Role:** Monitoring and enforcing environmental regulations.
- **Activities:** Conducting water quality assessments, regulating industrial discharge.

Effluent Treatment Plants (ETPs) and Sewage Treatment Plants (STPs)

- **Purpose:** Treating industrial waste and urban sewage before discharge into water bodies.
- **Challenges:** Ensuring effective operation and maintenance.

Public Awareness and Participation

- **Campaigns:** Educating the public on the importance of clean water.
- **Community Involvement:** Encouraging community-based monitoring and river cleaning activities.

(*https://en.wikipedia.org/wiki/List_of_rivers_of_India

https://en.wikipedia.org/wiki/List_of_major_rivers_of_India

https://en.wikipedia.org/wiki/Water_pollution_in_India

https://en.wikipedia.org/wiki/Water_pollution)

Water temperature is crucial, significantly impacting biological activity and growth. It affects water chemistry, influences water quantity measurements, and determines the types of organisms that inhabit water bodies. ((https://www.usgs.gov/special-topics/water-science-school/science/temperature-and-water)

Dissolved oxygen (DO) refers to the amount of oxygen present in water. Water bodies gain oxygen from the atmosphere and from aquatic plants. Flowing water, like that in a fast-moving stream, dissolves more oxygen than the still water in ponds or lakes. DO measures the oxygen available to aquatic organisms. The levels of dissolved oxygen in a stream or lake provide valuable information about its water quality. (https://www.epa.gov/national-aquatic-resource-surveys/indicators-dissolved-oxygen)

(https://www.usgs.gov/special-topics/water-science-school/science/dissolved-oxygen-and-water)

Elevated pH levels in drinking water can impart a bitter taste and lead to the buildup of deposits in water pipes and other appliances. Additionally, high pH reduces the effectiveness of chlorine as a disinfectant. Moreover, an elevated pH can cause metals in water to corrode or dissolve, increasing toxicity levels. Extreme pH levels can be harmful to aquatic organisms, as both excessively high and low pH can be lethal. pH also influences the solubility and toxicity of chemicals and heavy metals in water. Most aquatic species thrive in a pH range

of 6.5-9.0, although some can tolerate conditions outside this range. pH, which measures how acidic or basic water is, ranges from 0 to 14, with 7 being neutral. A pH below 7 indicates acidity, while a pH above 7 indicates a basic (alkaline) environment. pH is a critical parameter for assessing water quality. (https://atlas-scientific.com/blog/how-does-ph-affect-water-quality/) (https://www.usgs.gov/special-topics/water-science-school/science/ph-and-water)

Biochemical oxygen demand (BOD) measures the amount of dissolved oxygen consumed by aerobic bacteria to decompose organic material in water over a specific time, typically 5 days at 20°C. Expressed in milligrams of oxygen per litter, BOD indicates the level of organic pollution in water. Unlike chemical oxygen demand (COD), which measures all oxidizable substances, BOD specifically gauges biologically oxidized organic matter. Though water contains only a small amount of dissolved oxygen, this oxygen is crucial for aquatic life. (https://en.wikipedia.org/wiki/Biochemical_oxygen_demand) (https://www.usgs.gov/special-topics/water-science-school/science/biochemical-oxygen-demand-bod-and-water)

Excessive nitrate consumption can impair the blood's oxygen-carrying capacity, leading to methemoglobinemia, or "blue baby syndrome," which is particularly dangerous for bottle-fed infants under six months. Symptoms include bluish skin, low blood pressure, increased heart rate, headaches, stomach cramps, and vomiting. People with conditions like anaemia, cardiovascular or lung disease, sepsis, and certain metabolic disorders are more vulnerable. Recent studies suggest potential links between nitrate/nitrite exposure and health issues like increased heart rate, nausea, headaches, abdominal cramps, and possibly an increased risk of cancer, particularly gastric cancer, though more research is needed. (https://www.health.state.mn.us/communities/environment/water/contaminants/nitrate.html)

Water is crucial for life, yet many people lack access to clean drinking water, leading to waterborne infections. Microbial pollution from sources like faecal matter, hospitals, industries, and cattle farms increases bacterial levels in water bodies. Coliform bacteria, especially Escherichia coli, are key indicators of faecal contamination. The multiple tube fermentation technique traditionally detects coliforms by observing acid and gas production from lactose fermentation, determining water potability based on the most probable number index value (MPN/100 ml). Faecal streptococci and Clostridium perfringens are also used as faecal pollution indicators, confirmed via esculin

hydrolysis or catalase-negativity and sulphite reduction tests. Modern methods like PCR and enzymatic assays offer rapid detection of these indicators and other enteric pathogens. Additionally, the standard plate count (SPC) and biochemical oxygen demand (BOD) measure bacterial and organic pollution in water. Bacteriological analyses reveal that water polluted by sewage is unsafe for drinking and recreation, highlighting a major challenge of ensuring safe water access in the 21st century. (https://www.sciencedirect.com/science/article/pii/S2772427121000085)

4. **Research methodology**:

4.1 Research design

Sample datasets were collected from NITI Aayog's website for rivers, minor rivers, lakes; ponds; tanks, canals; drains and ground water. Out of two datasets made available there, source datasets were used to draw insights about water quality for different main water tributaries, their minor branches, other water bodies like tanks, canals, drains etc. Quality of groundwater was analyzed state wise.

4.2 Sampling technique

A mix of several sampling techniques is used for this water quality analysis project. **Simple Random Sampling is employed at time to draw insights from entire data without dividing it into groups. Then datasets were again analysed with conditional water body (river name, minor river, canal etc.) and geographical filters (state wise analysis): stratified sampling and cluster sampling, to draw insights for individual state and also compared with other state.**

4.3 Data collection

All the datasets used in this project are collected from NDAP (National Data Analytic Platform), NITI Aayog.

4.4 Data preparation

To analyze data for valuable insights,

- First, the datasets were opened, viewed and generally analyzed in MS Excel to plan for further steps of advanced cleaning and filtering.
- Then, the datasets were cleaned using python jupyter notebook and analyzed to draw helpful visualizations. Some charts and graphs were made with Power BI.

4.5 Data analysis

Trends in data of datasets

Here we will try to analyze the trends in pollution level of rivers in India based on factors like temperature of water, dissolved oxygen, pH, conductivity level, BOD of water, Nitrate and Nitrite required for water, Fecal coliforms and coliforms present in it.

4.5.1 RIVERS IN INDIA

India, home to a vast network of rivers, faces significant challenges in maintaining water quality. Major rivers such as the Ganga, Yamuna, Godavari, Krishna, and Cauvery are vital lifelines for millions, providing water for drinking, agriculture, and industry. However, these rivers are severely polluted due to industrial effluents, untreated sewage, and agricultural runoff. The Ganga and Yamuna, among the most sacred rivers, suffer from high levels of biological oxygen demand (BOD), faecal coliforms, and toxic heavy metals. Industrial discharge in rivers like the Godavari and Krishna introduces hazardous chemicals, while agricultural runoff in the Cauvery contributes to nutrient pollution. Despite various government initiatives like the National Mission for Clean Ganga (NMCG), the lack of adequate sewage treatment infrastructure and enforcement of pollution control regulations exacerbates the situation. Effective management of these rivers is crucial for ensuring sustainable water resources and protecting public health in India.

4.5.1(a). Temperature across different rivers

Water temperature not only affects its chemistry but also determines the types of organisms that can inhabit a particular water body within a certain temperature range. High temperatures in river waters indicate pollution.

From the collected dataset, rough visualization of river temperature distribution shows a range from approximately 0°C to 50°C (fig. 2), with minimum temperatures spanning from around 0°C to 35°C and maximum temperatures from around 0°C to 50°C(fig.1).

**For all datasets, NITI made two values available for each factor (temperature, dissolved oxygen, BOD, etc.): the minimum value and the maximum value recorded for each sample. Visualizations of one or both values—minimum recorded value and maximum recorded value—are included where necessary for illustration.
Temperatures were measured in °C, dissolved oxygen in mg/L, conductivity in µS/cm, BOD in mg/L, nitrates in mg/L, and faecal coliforms and coliforms in MPN/100mL.

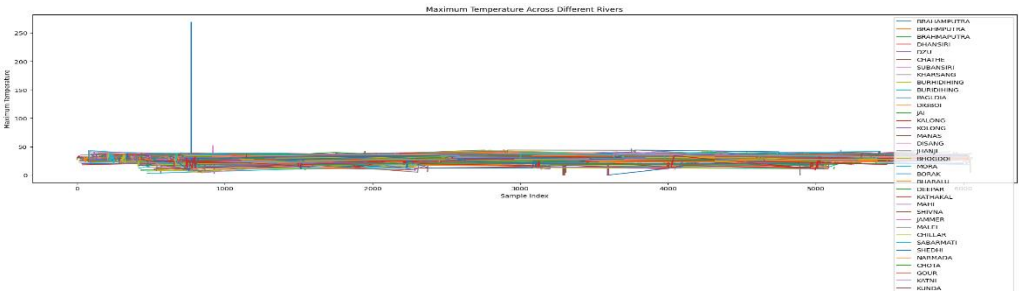


Fig. 1 India has around 400 rivers. The temperature (generally high) of Indian river waters, which is one indication of pollution, is represented for different rivers against a sample axis. Each colour represents a different river. The X-axis is the sample axis and the Y-axis represents temperature in °C. The list on the right includes approximately 450 water bodies, trimmed to save space and maintain clarity.

4.5.1(b). Dissolved oxygen across different rivers

Dissolved oxygen adequate for taste and human health should be above 6mg/L. To support marine organisms, it should be at least 5mg/L (check in study hypothesis for standards).

In the collected dataset for rivers, most samples had dissolved oxygen between ~0mg\L - ~10mg\L(fig.2).

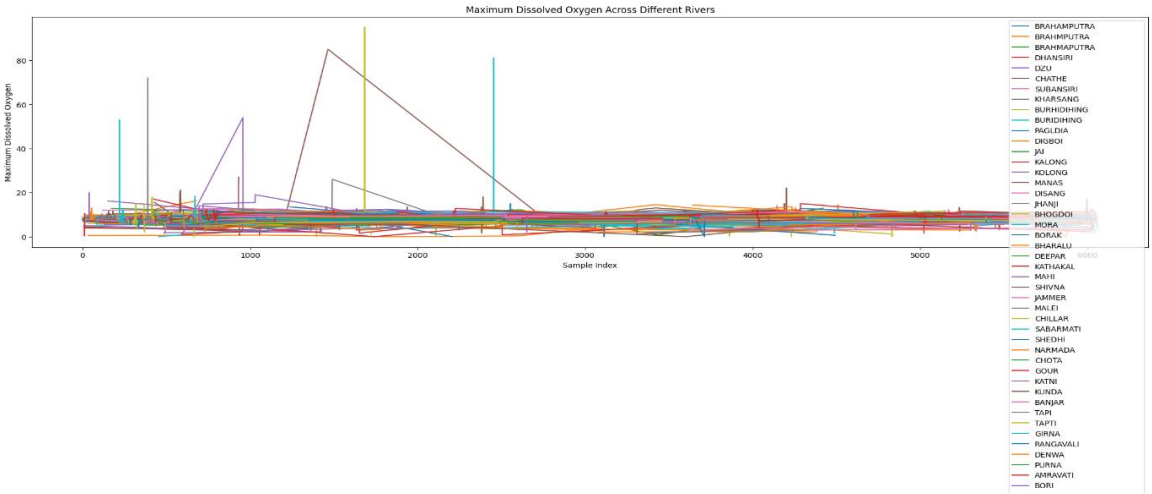


Fig. 2 Maximum dissolved oxygen across different rivers. X-axis is a sample axis and y-axis represent dissolved oxygen(mg/L) for different Indian rivers. Each colour is for a different river. The list on the right is of approx. 450 water bodies. It is trimmed to save space and maintain clarity.

4.5.1(c). pH of water across rivers

On an average, pH is close to standard 6mg\L value for most river samples. Only some markings show high pH – acidic nature.

4.5.1(d). Conductivity of water across rivers

Both the minimum and maximum values of conductivity were found to be close to less than 500µS/cm which is suitable for drinking purpose.

4.5.1(e). BOD (Biochemical oxygen demand) across rivers

Out of 6061 samples collected for river waters between 2012-2019, 1806 showed minimum BOD levels greater than 2 mg/L (fig. 3a) (water above this level is not suitable for human drinking purposes), and around 367 samples showed BOD levels greater than 5 mg/L (fig. 3b) (above this level, water does not sustain life well and indicates pollution). (fig. 3)

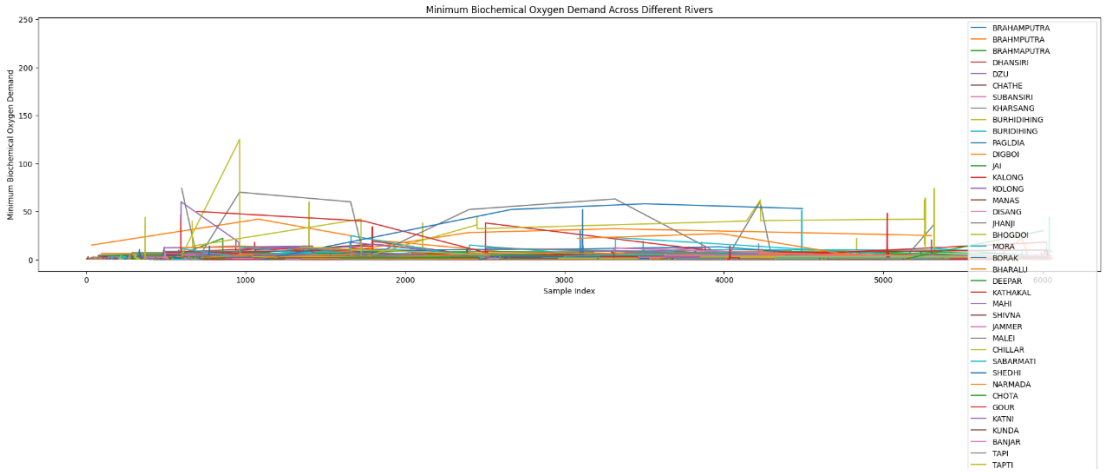


Fig. 3(a)

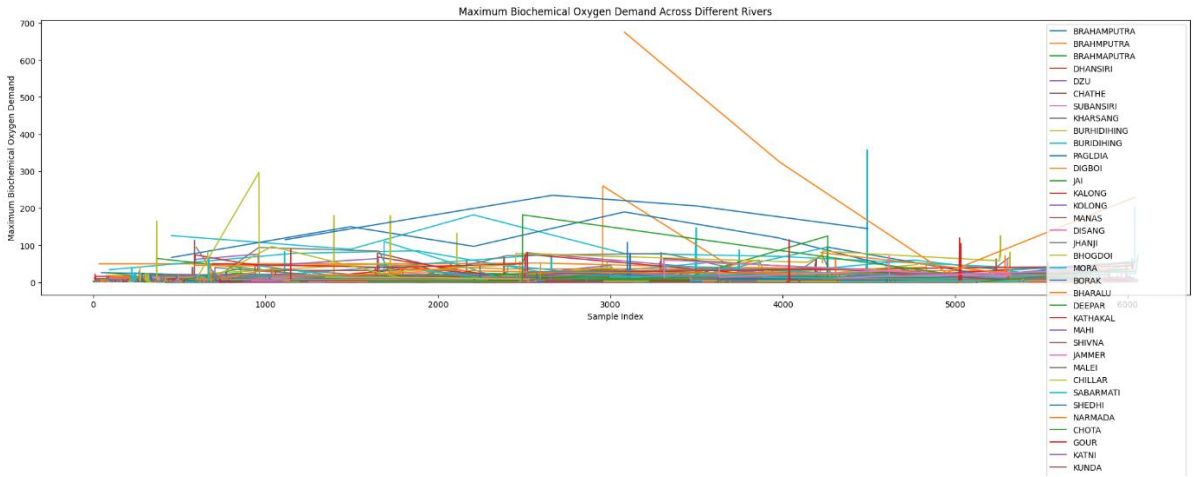


Fig. 3(b)

Fig. 3 Biochemical oxygen demand (BOD) indicates the amount of dissolved oxygen (DO) consumed by aerobic bacteria breaking down organic material in a water sample over a specific time and temperature. It often serves as an indicator of the level of organic pollution in water. In the figure, BOD levels of over 400 river waters from 6061 samples are shown along a sample axis, with each colour representing a different river. On the right, there is a list of around 450 names. The upper graph represents the minimum BOD levels observed, while the lower one indicates the maximum BOD levels recorded.

4.5.1(f). Nitrate N + Nitrite N required

Presence of nitrates and nitrites was found to be minimal, close to ~0mg\L in the visualizations of the collected dataset for most samples.

4.5.1(g). Fecal coliforms and Coliforms in water

Around ~5000 samples out of 6061 show presence of some coliforms and ~2000 samples had coliforms greater than 200CFU\100mL which indicates that the water of those 2000 bodies was highly polluted. (fig. 4)

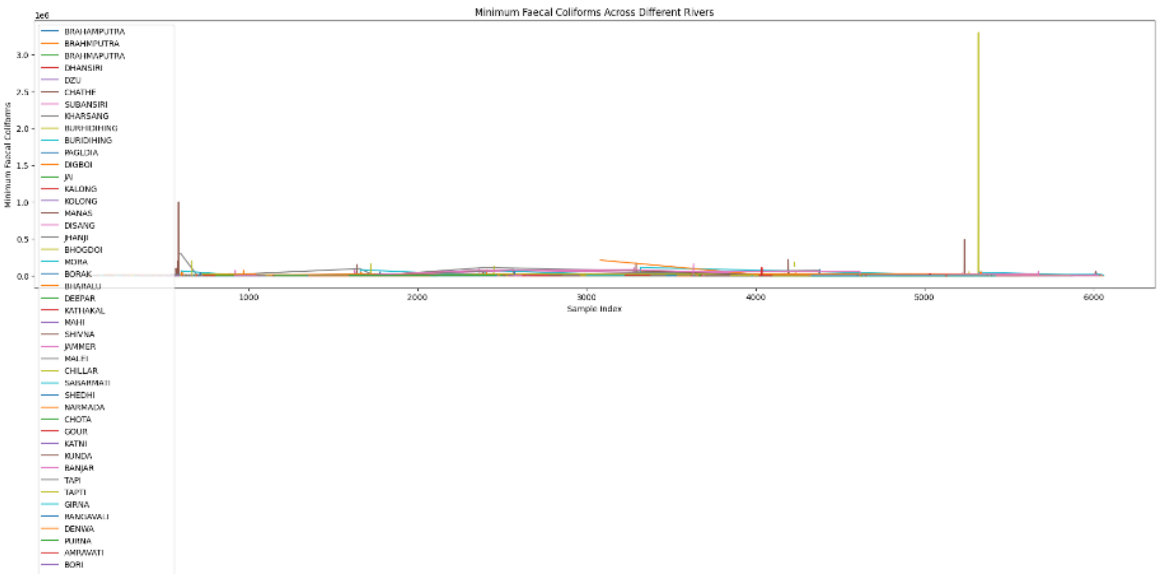


Fig. 4(a)

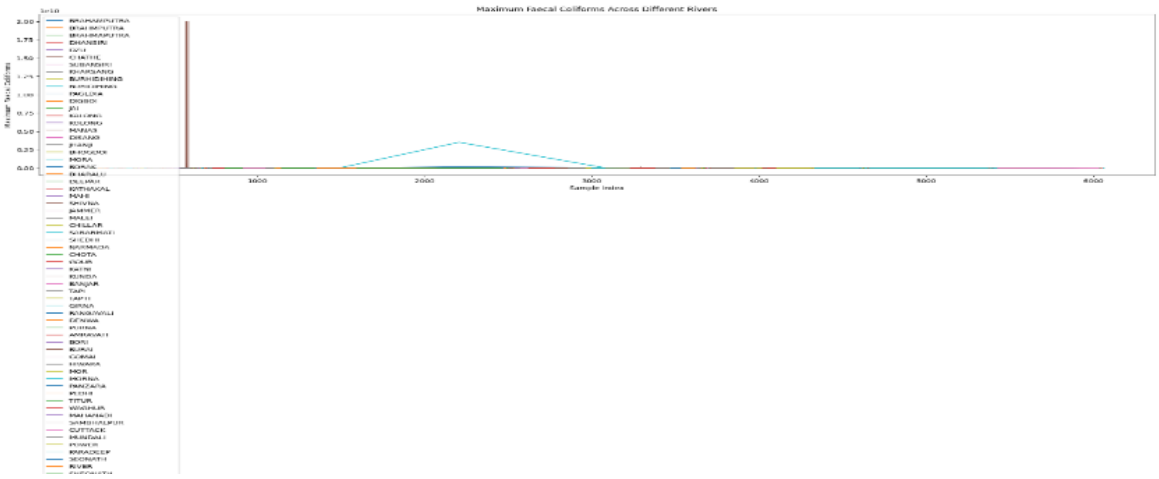


Fig. 4(b)

Fig. 4 Figure shows the distribution of fecal coliforms in river waters. X-axis is a sample axis and y-axis represent fecal coliforms in MPN/100mL for different rivers. List of water bodies on right contains more than 450 names. They are graphed based on 6061 river water samples collected between 2012-2019(list is trimmed to save space)

4.5.2. MEDIUM, MINOR RIVERS

In the dataset of 2198 samples collected for medium, minor rivers 1333 had minimum dissolved oxygen less than 6mg\L (minimum 6mg\L is required for safe drinking water and minimum 5mg\L for life in water) and 860 had less than 5mg\L (below the requirement necessary to sustain marine life). 545 had pH values less than 6.5(acidic) and 239 had more than 8.5(basic). 344 had minimum conductivity greater than 500µS/cm(conductivity>500 µS/cm is not suitable for drinking – check study hypothesis for standards), 6samples showed minimum conductivity greater than 30000 µS/cm (above the limit which sustains marine life) and 104 had maximum conductivity greater than 50000 µS/cm (unfit for life in water). 656 had maximum BOD greater than 5mg\L (unsuitable for marine life), 625 had minimum BOD greater than 2mg\L (less than 2mg\L is required for drinkable water). 1824 showed presence of some fecal coliform. 436 had minimum fecal coliform greater than 200CFU\100mL (unfit for aquatic life) 2002 had some coliforms in them and 1586 had maximum fecal coliform greater than 200CFU\mL.

4.5.3 LAKES, PONDS, TANKS

In the collected dataset for 2799 samples between 2012-2020, Karnataka had maximum number of samples of water bodies (lakes, ponds, tanks) ~619(see in fig. 5) samples, with maximum number of lakes and tanks (~245, ~367) samples with 5-6 other water bodies. States like Telangana, Assam, Madhya Pradesh, Kerala, Gujarat, Rajasthan, Andhra Pradesh, Manipur, Jammu and Kashmir, Orisha, Tripura, west Bengal too have good number of samples. Telangana having ~136lakes, ~55ponds and ~123tanks samples, Assam have ~29lakes, ~130ponds, ~5tanks samples; Madhya Pradesh ~21lakes, 12ponds samples; Kerala ~79lakes, ~10ponds samples; Gujarat ~94lakes, ~10ponds, ~5tanks samples; Rajasthan ~80lakes, ~4-5ponds, ~50other samples; Andhra Pradesh ~10-12lakes samples; Manipur ~25lakes, ~65ponds samples; Jammu and Kashmir ~130lakes samples; Orisha ~30lakes, ~35ponds samples; Tripura ~24lakes, ~40ponds samples; West Bengal ~56lakes samples (fig. 5)

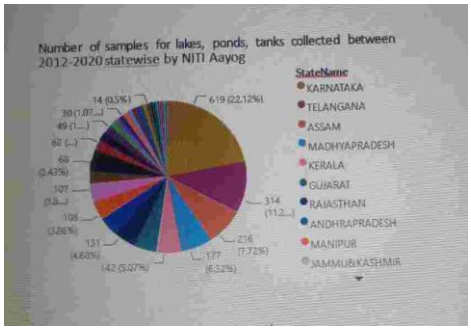


Fig. 5 Number of samples for lakes, ponds and tanks collected between 2012-2020 state-wise by NITI Aayog

4.5.3(a). Temperature range

Lakes minimum temperature ranges between ~0°C - ~35°C (fig.6a) and maximum reaches up to around ~40°C (fig. 6b), temperature of tanks is maintained between ~15°C - ~35°C, temperature of ponds(green line in graph) varies from ~15°C - ~35°C with some markings below 15°C and above 35°C which can be due to extreme cold (fig. 6a) in winter and sun (fig. 6b) in summer. (fig. 6)

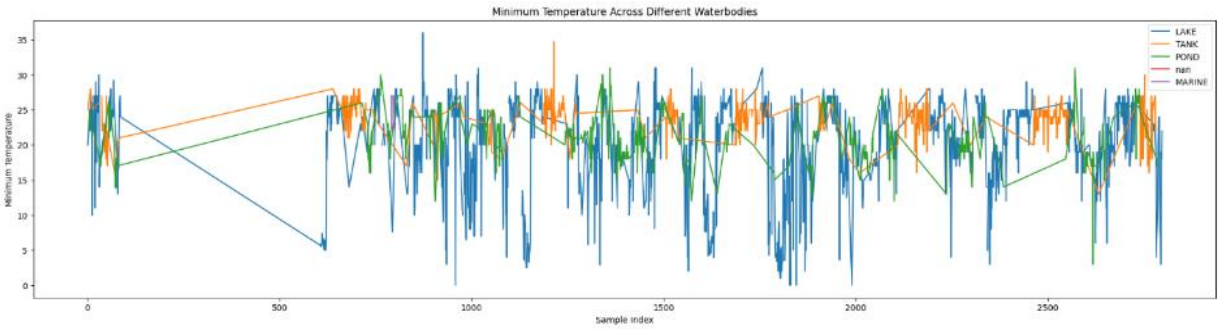


Fig. 6(a)

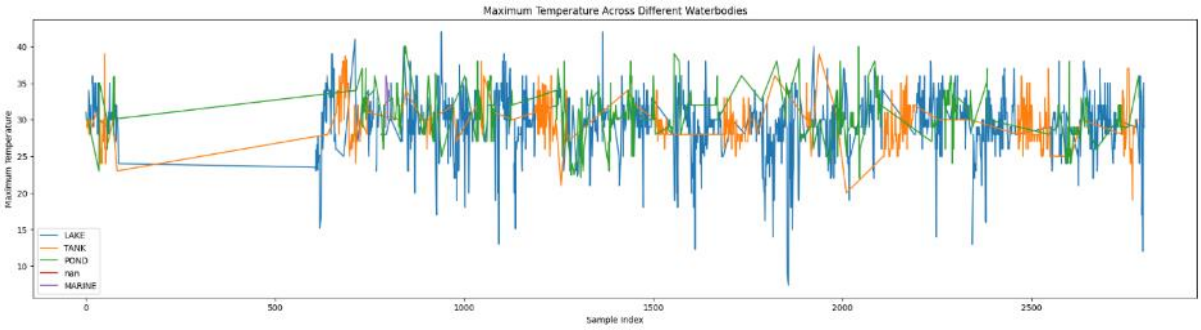


Fig. 6(b)

Fig. 6 Figure shows the comparative analysis graph of the temperature distribution of different water bodies (lakes, ponds, tanks and marine). Each color is for a different body- blue is for lake, orange is for tank, green is for pond. X-axis is a sample axis and y-axis show temperatures in °C. 6(a) shows minimum temperatures recorded and fig 6(b) shows maximum temperatures recorded

4.5.3(b). Dissolved oxygen

Dissolved oxygen in lakes (blue line in graphs) is ~5-6mg\L (which is a good measure for drinking water and also suitable for aquatic life) with some below minimum ~5mg\L readings. Also dissolved oxygen for tanks and ponds show similar readings. (fig. 7)

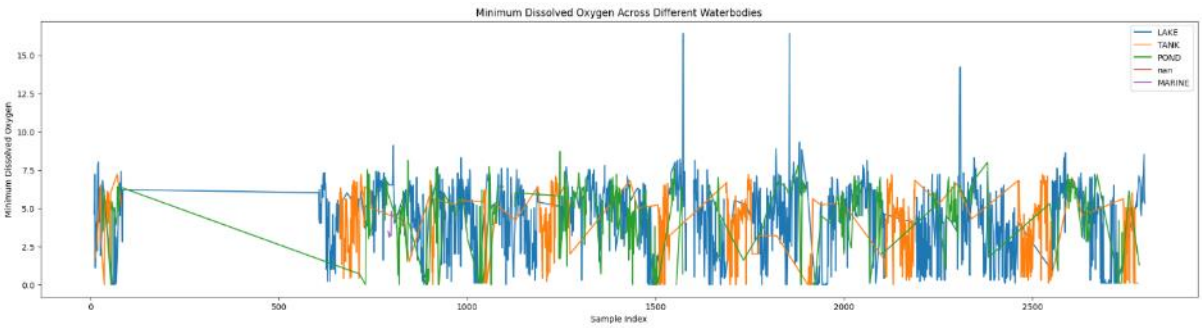


Fig. 7(a)

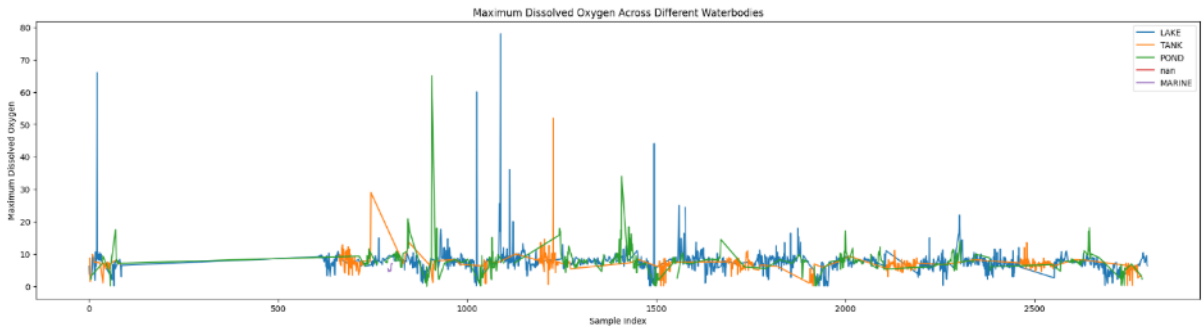


Fig. 7(b)

Fig. 7 Figure shows dissolved oxygen(mg/L) for different water bodies. X-axis is a sample axis and y-axis represent dissolved oxygen in mg/L. (these visualizations are for 2799 samples collected between 2012-2020 for lakes, tanks, ponds etc. by NITI Aayog)

4.5.3(c). pH values

pH of all three water bodies (lakes, ponds, tanks) as per data is ~6.5 - ~8.5(6.5 – 7.5 is good for drinking purpose and 7.5 – 8.5 pH range is typical for marine life - which is good for underwater life – see study hypothesis), with some readings showing the data of an extremely acidic lake. (fig. 8)

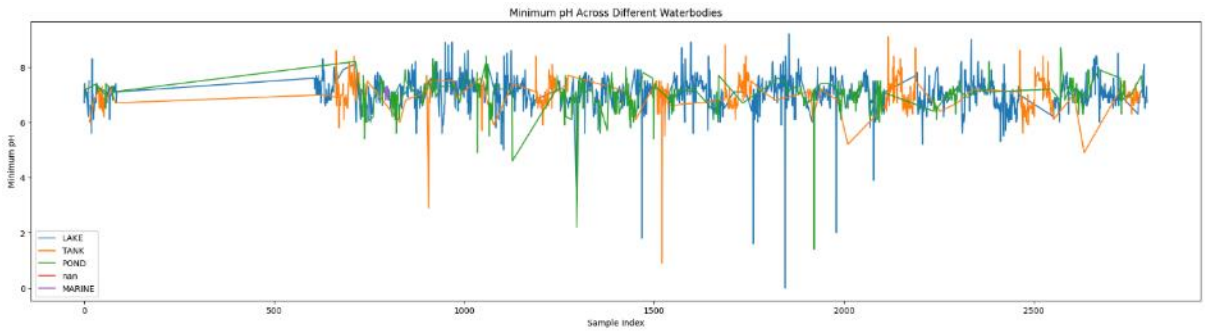


Fig. 8(a)

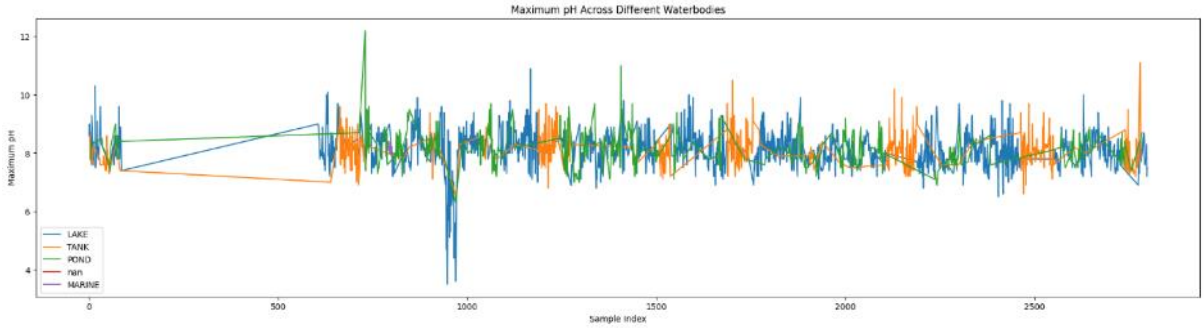


Fig. 8(b)

Fig. 8 Figure shows the pH ranges for different water bodies. X-axis is a sample axis and y-axis represent pH for water bodies

4.5.3(d). Conductivity levels

High conductivity levels, above ~50000µS/cm are indicators of pollution and the presence of chlorides, phosphates and nitrite ions in water. 6lake samples

in 2016 and 2017 and 2-3samples in 2018-2020 shows conductivity above 50,000µS\cm indicating high pollution in those 6 water bodies. (fig. 9)

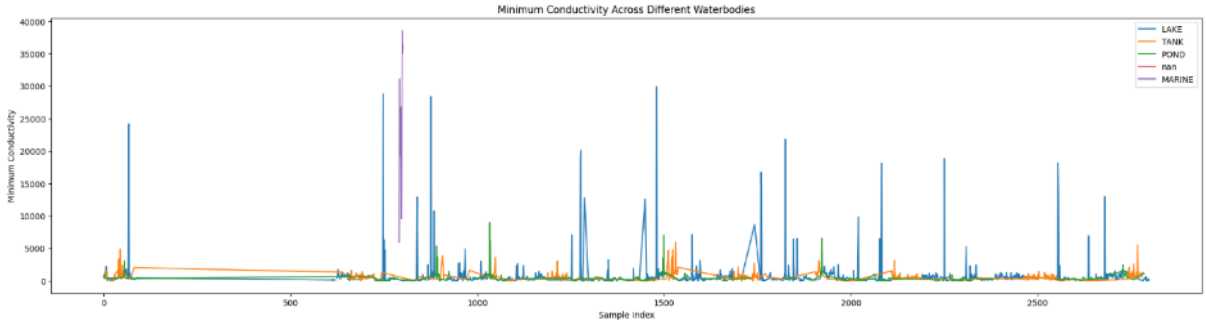


Fig. 9(a)

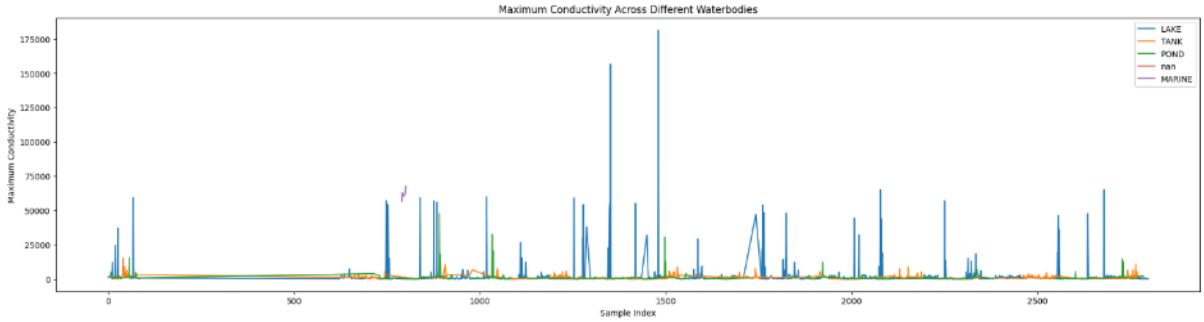


Fig. 9(b)

Fig. 9 Figure for conductivity levels of lakes, ponds and tanks. X-axis is a sample axis and y-axis represent conductivity levels in µS/cm

4.5.3(e). Biological Oxygen Demand

For water to be useful for humans it's BOD should be less than 2-5mg\L (2mg\L is good for drinking purpose for humans and BOD less than 5mg\L sustains marine life). Some readings of lakes and ponds show high pollution. (fig. 10)

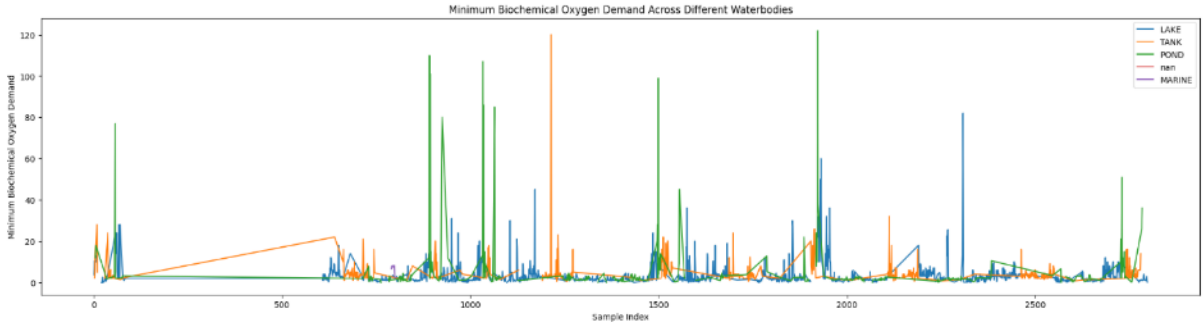


Fig. 10(a)

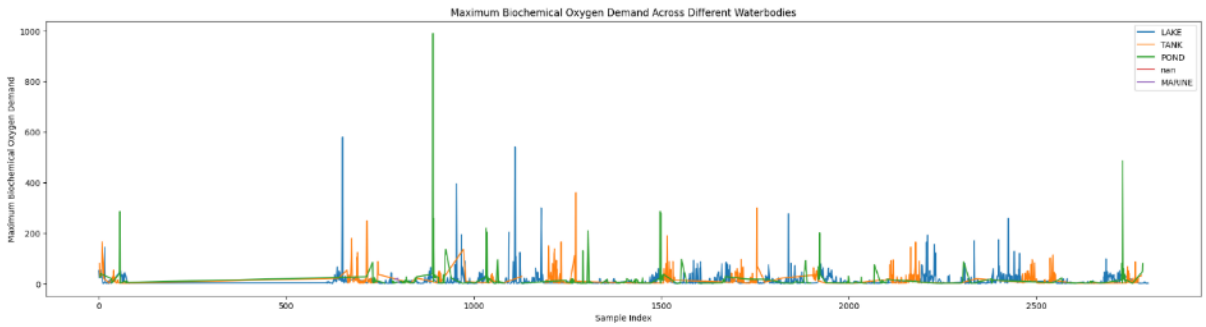


Fig. 10(b)

Fig. 10 X-axis is a sample axis and y-axis represent BOD (biological oxygen demand) in mg/L. Figure 10a is for minimum BODS recorded and fig. 10b shows maximum BODs recorded for samples of water bodies (lakes, ponds etc.) collected between 2012-2020

4.5.3(f). Nitrates and Nitrites present

As per standards nitrates, nitrites presence should be less than 1mg\L for humans and less than .1mg\L can only make life in water possible.

Data shows presence of nitrites in tanks (orange). Out of tank data samples collected between 2016-2020 125 samples showed minimum nitrites presence greater than 1mg\L. (fig. 11). Some lakes also show high presence of nitrates (fig. 11 blue line)

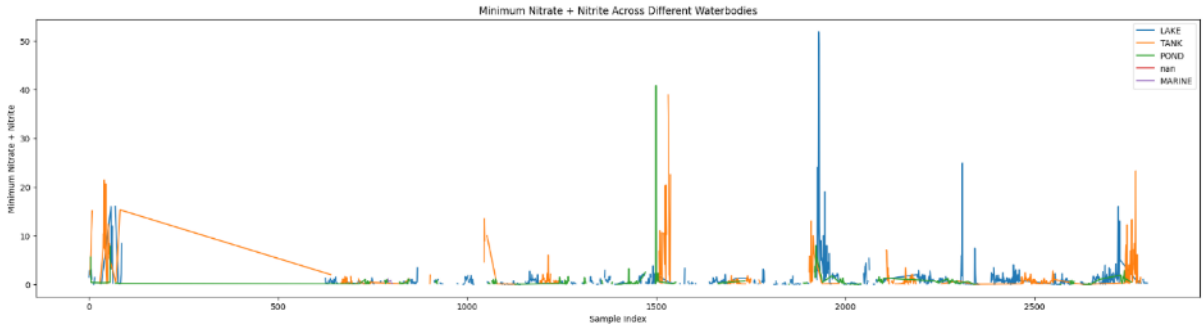


Fig. 11(a)

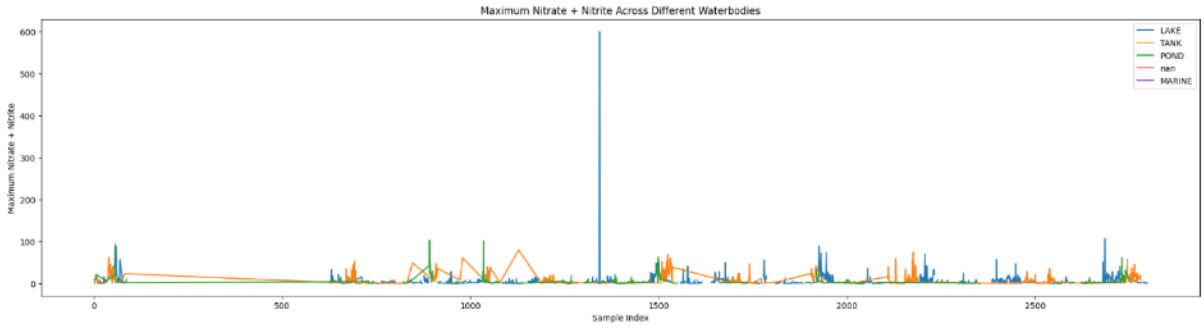


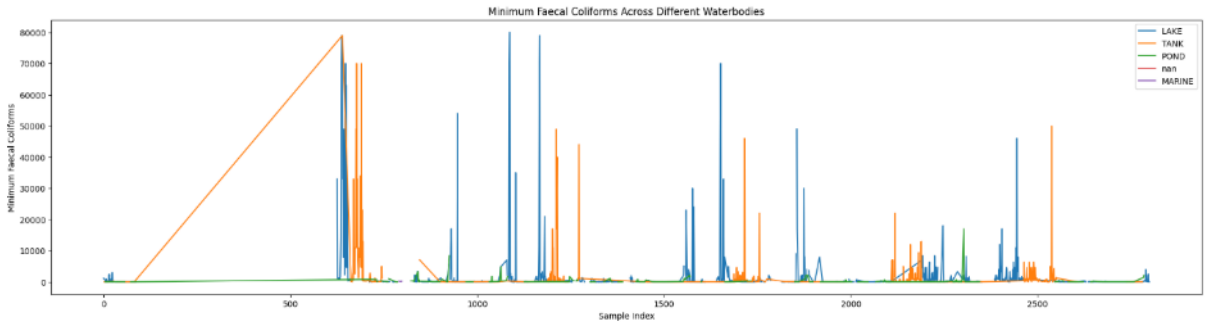
Fig. 11(b)

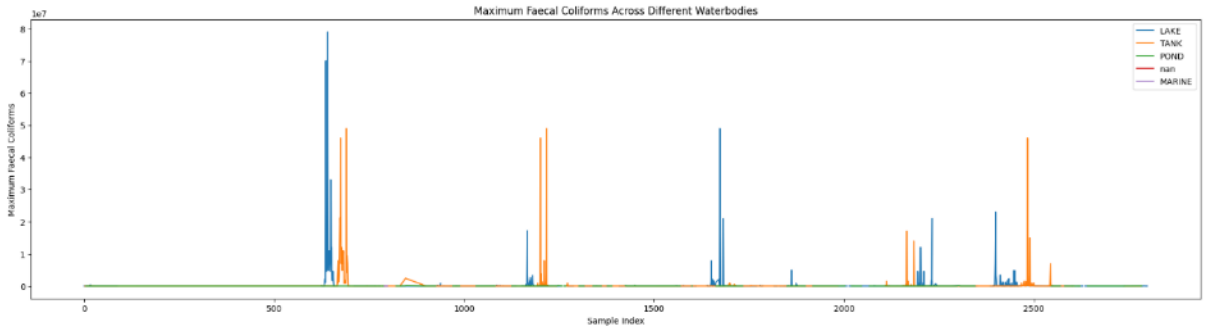
Fig. 11 X-axis is a sample axis and y-axis represent nitrates and nitrites present in mg/L for different water bodies.

4.5.3(g). Fecal coliforms and coliforms present

Ideally drinking water should be free from fecal coliforms and coliforms but aquatic beings can survive under 200CFU(MPN)/100Ml.

Many lake samples show that their water wasn't drinkable, showing high pollution. (fig. 12) (blue – lake, orange – tank)





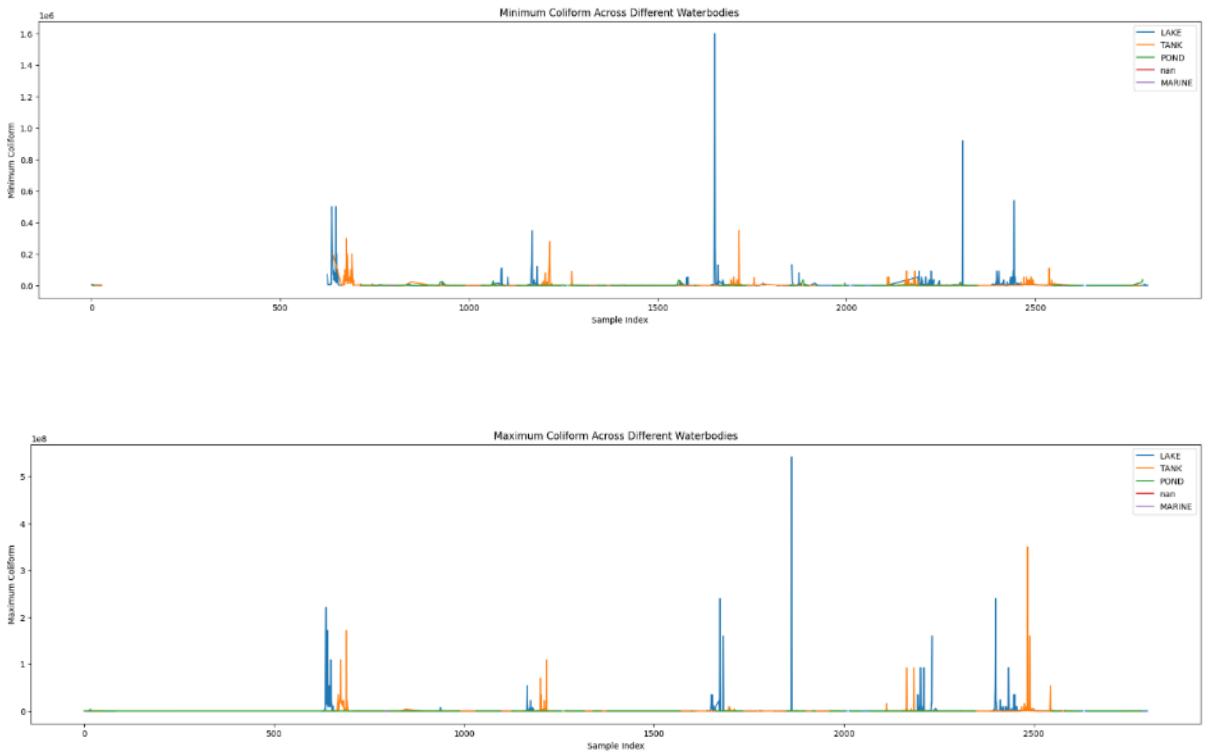


Fig. 12 X-axis is a sample axis and y-axis represent fecal coliforms (top 2 graphs) and coliforms (bottom 2 graphs) in MPN/100mL

4.5.4. CANALS, SEAWATER, DRAINS, STP and WTP

Visualizations for 915 samples collected between 2017-2019 from canals, drains, sewage treatment plants, creeks, and marine areas show high conductivity values. BOD levels for sewage treatment plants, drains, creeks, industrial drains, marine areas, and nallahs are high, indicating significant pollution. Nitrate levels are also elevated in most water bodies. Some readings for canals and creeks show a high presence of faecal matter.

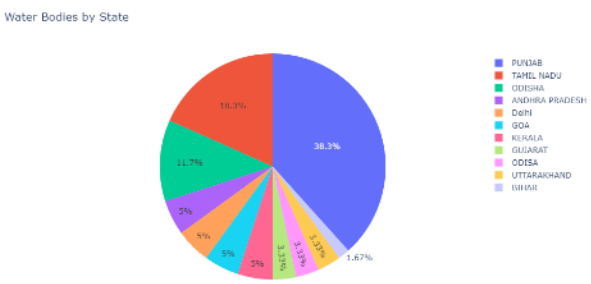


Fig. 13 Figure shows the number of samples collected for water treatment plants state-wise between 2017-2019 by NITI Aayog

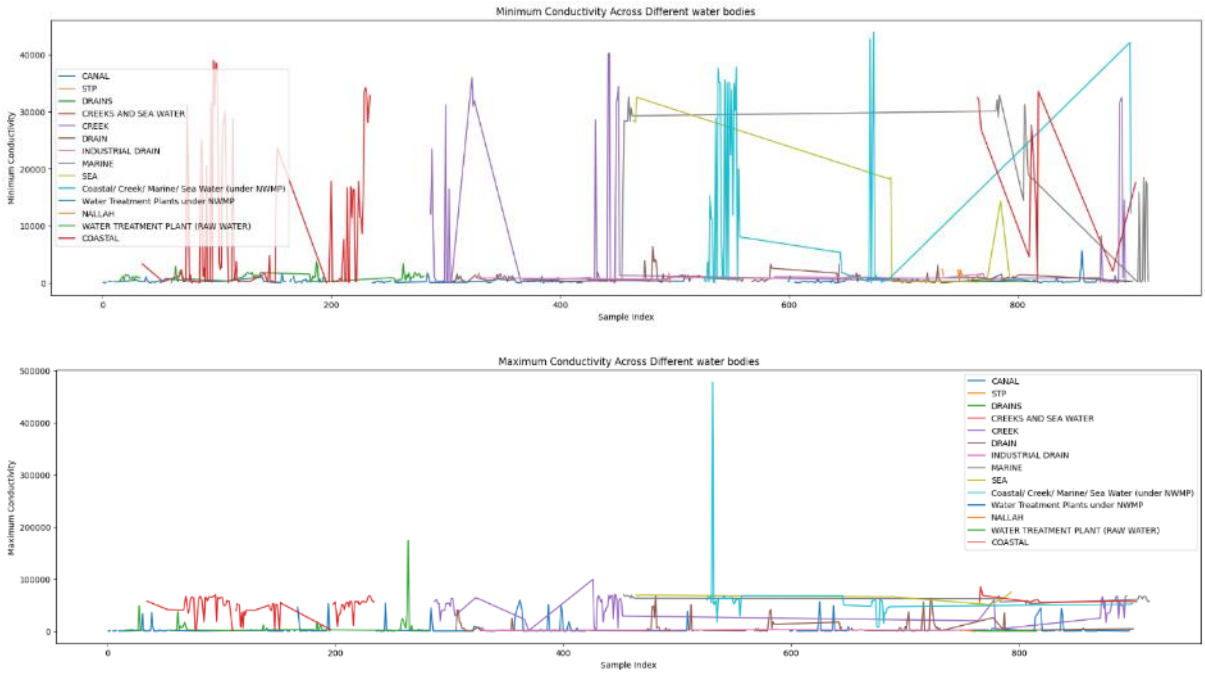
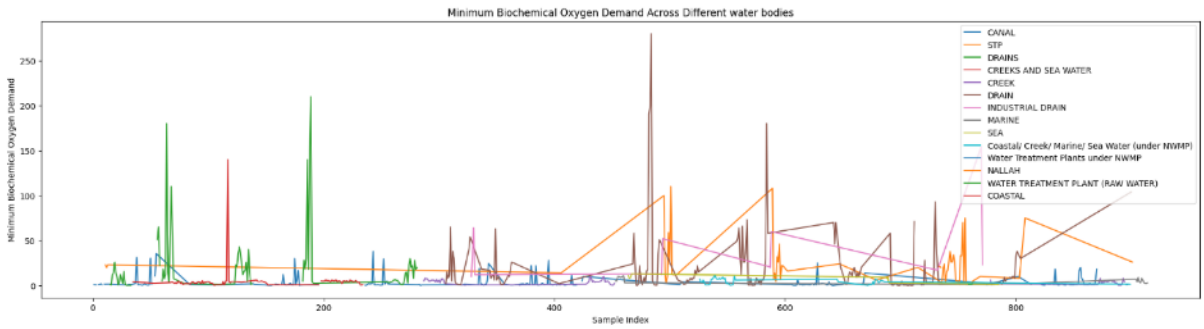


Fig. 14 X-axis is a sample axis and Y-axis represents conductivity in µS/cm for different water bodies



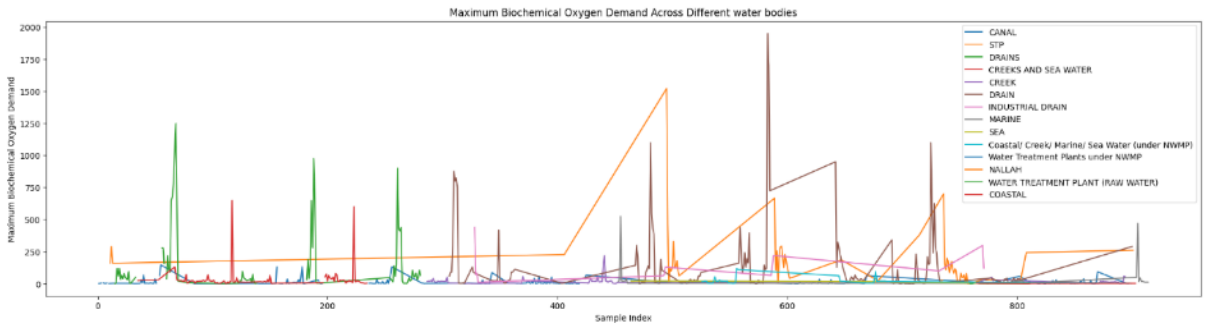


Fig. 15 X-axis is a sample axis and Y-axis represents BOD for different water bodies in mg/L

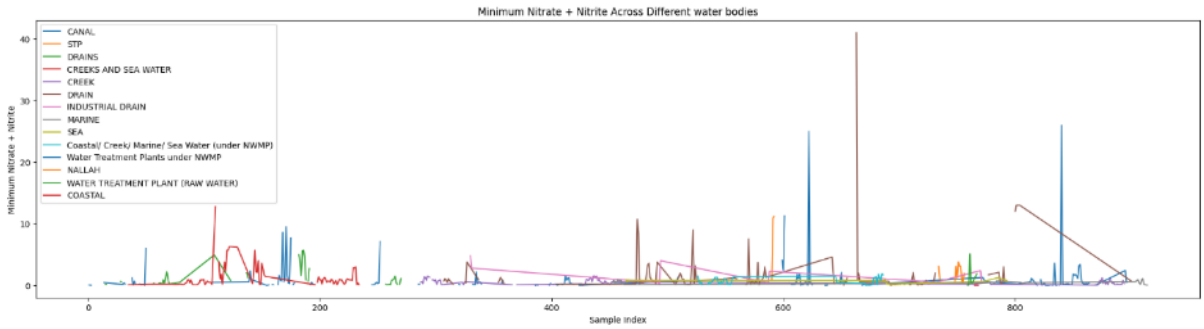
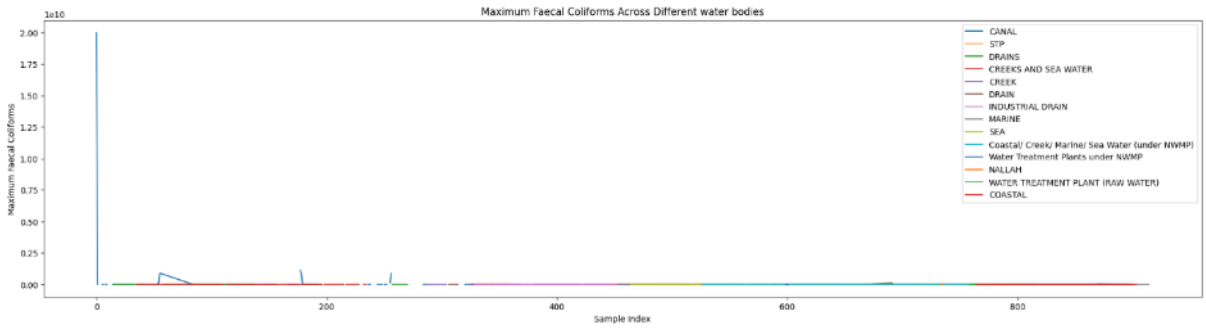


Fig. 16 X-axis is a sample axis and Y-axis represents nitrates and nitrites(mg/L) present in water samples of water bodies



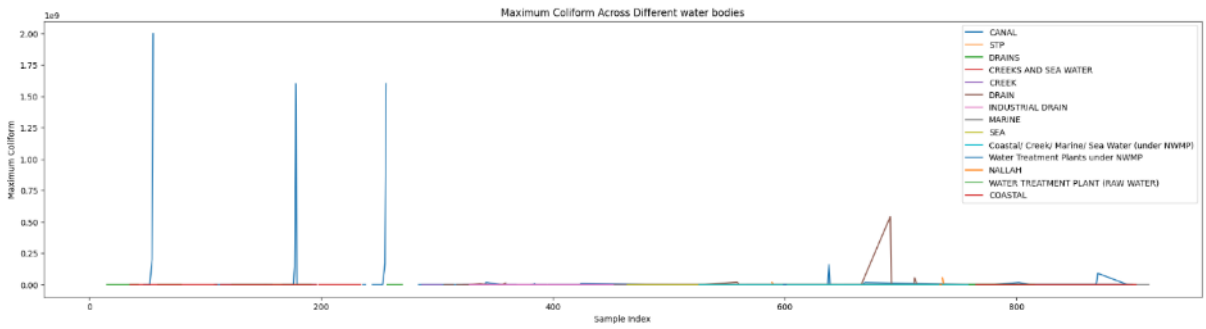


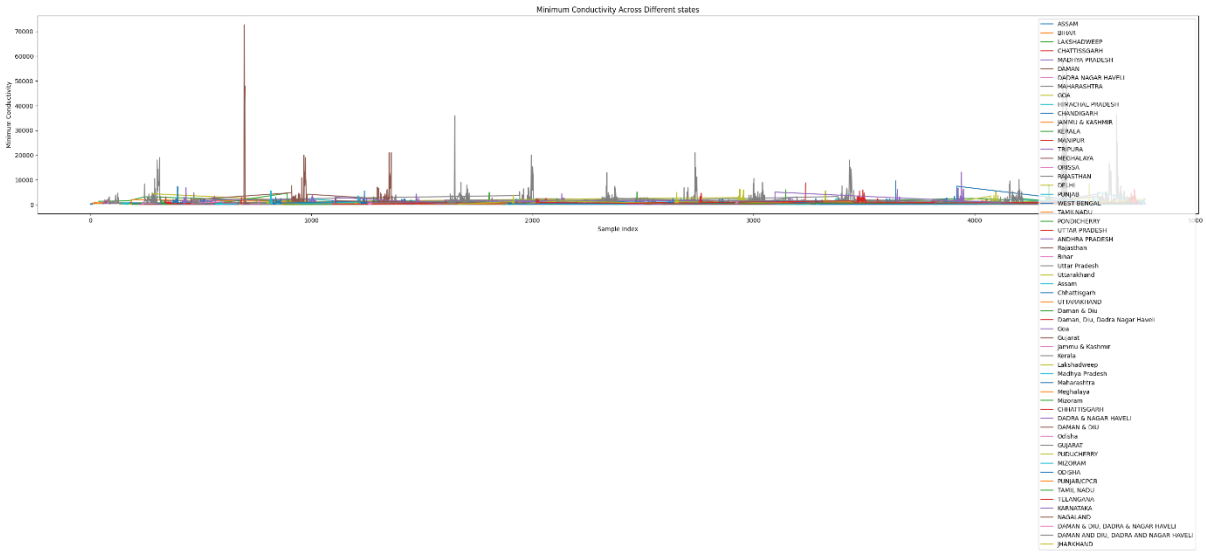
Fig. 17 X-axis is a sample axis and Y-axis represents fecal coliforms and coliforms present in MPN/100mL

4.5.5. GROUND WATER

Some readings of conductivity for ground water shows high values.

For most samples BOD is low and only few had high value.

Some readings show high value of nitrites and nitrites for ground water which can be due to the use of fertilizers for agricultural activities and discharges from industrial processes.



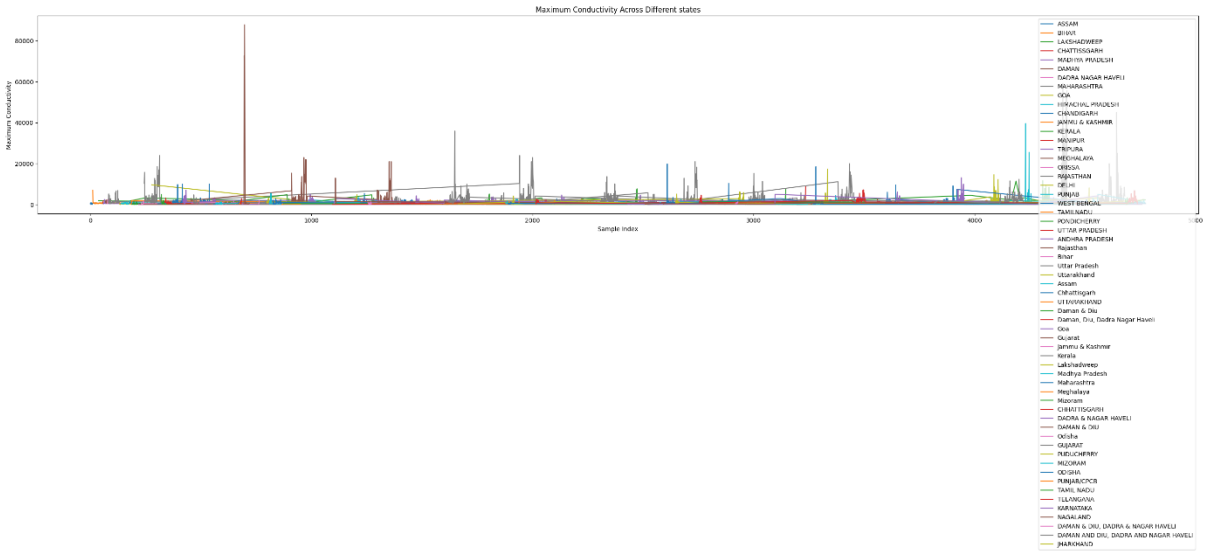


Fig. 18 Conductivity of water for drinking should be around 500µS/cm. X-axis is a sample axis and Y-axis represents conductivity for ground water in µS/cm state-wise

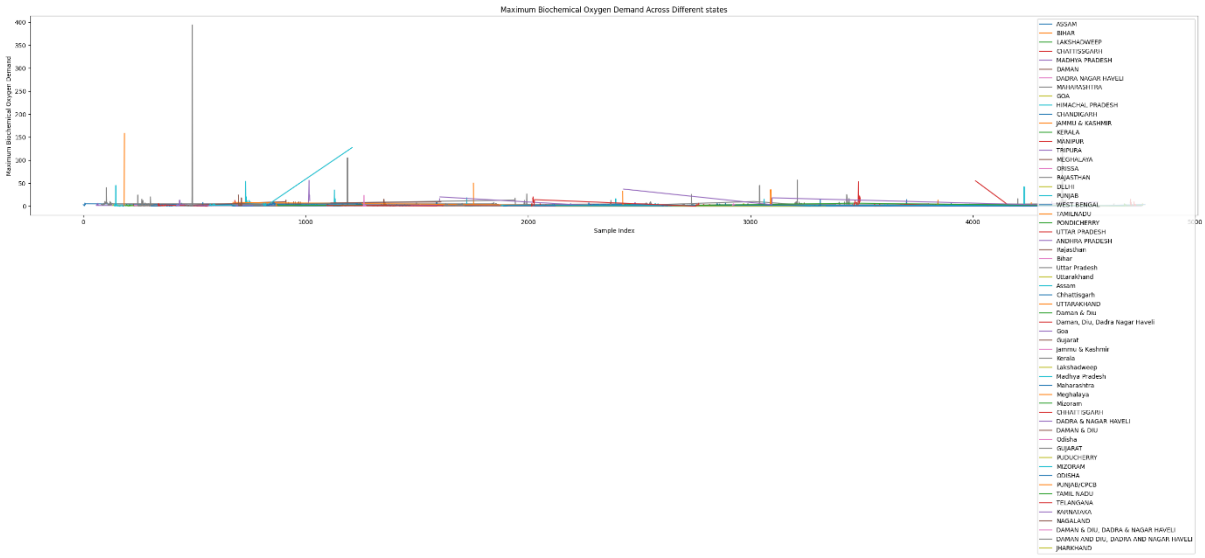


Fig. 19 Figure illustrates the biochemical oxygen demand(mg/L) for ground water for different states

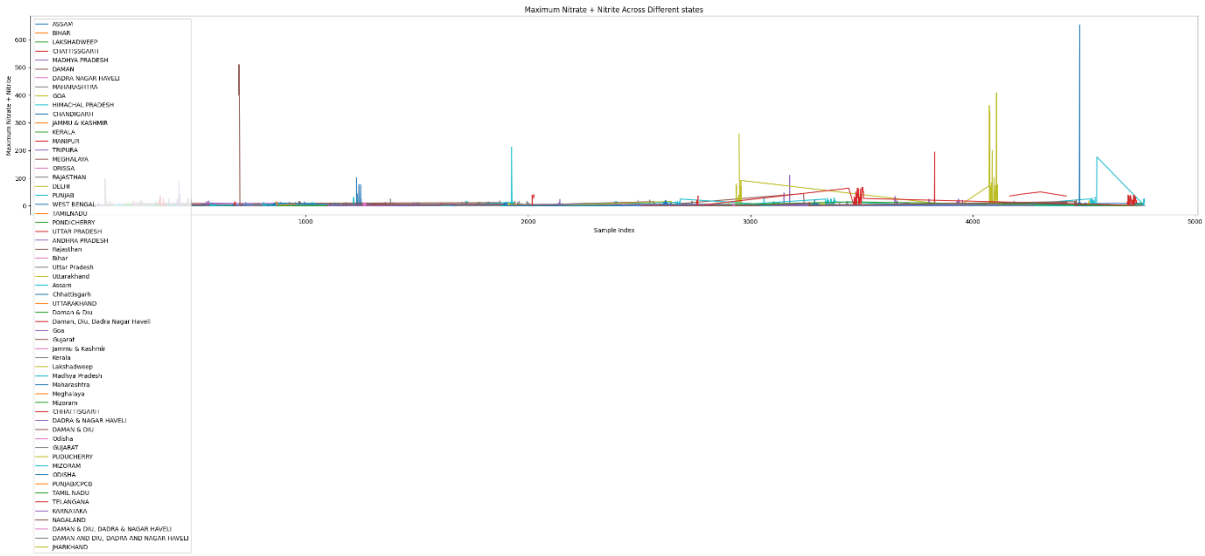


Fig. 20 Figure illustrates the presence of nitrates in ground water state-wise. For water to be suitable for drinking, ideally the presence of nitrates should be less than 1mg/L. Most reading in the figure show nitrate ions close to 0mg/L with some moderate or high peaks indicating pollution. (x-axis is the sample index and y-axis represent presence of nitrates in mg/L for Indian states)

4.6 Machine Learning-Based Analysis

To complement traditional statistical analysis, machine learning techniques were employed to uncover hidden patterns, predict water quality, and classify pollution levels across different water bodies in India.

4.6.1 Feature Selection and Preprocessing

The analysis utilized key physicochemical and biological parameters derived from the study hypothesis, including temperature, dissolved oxygen (DO), pH, conductivity, biochemical oxygen demand (BOD), nitrate concentration, faecal coliforms, and total coliforms.

All features were standardized using **z-score normalization** to ensure equal contribution during model training. Missing and extreme values were handled using filtering and clipping techniques to maintain numerical stability.

4.6.2 Clustering Analysis (Unsupervised Learning)

K-Means clustering revealed three distinct water-quality regimes within the dataset: a dominant moderately polluted cluster, a critically contaminated hotspot cluster, and an anomalous cluster likely caused by data inconsistencies. The persistence of these structures across training and testing datasets suggests stable underlying pollution patterns. However, the presence of physically implausible values highlights limitations in data preprocessing, emphasizing the need for stricter range validation, outlier treatment, and domain-informed

feature engineering to produce more realistic and scientifically robust environmental assessments.

- Number of clusters: 3
- Interpretation:

Plot shows:

Cluster 0 (major dense group)

Large concentration near origin.

This means:

- Most water samples share similar characteristics
- Represents the **dominant water quality pattern**

Likely:

- Moderately contaminated / average-quality water

This cluster likely represents:

Most Indian surface water bodies experiencing:

- chronic moderate contamination
- urban runoff
- untreated sewage influence
-

Cluster 1 (small separated group)

A few points separated vertically.

This indicates:

- Rare but distinct water samples
- Different pollution chemistry

Likely:

- unusual chemical composition
- nitrate-heavy or measurement anomalies

• pH = **2245** • nitrate = **81775** This strongly indicates: **data corruption / unit mismatch / parsing error**

Cluster 2 (far isolated points)

Very distant from all others.

This is extremely important.

It indicates:

Extreme outlier pollution events

These are likely severely contaminated water samples.

Train:

- DO = **2.4**
- BOD = **34.33**
- faecal coliform = **2×10^{10}**
- total coliform = **2×10^{10}**

Test:

- DO = **3.4**
- BOD = **91.95**
- coliform = **10^{10} scale**

This is severe contamination.

Interpretation:

These samples represent:

Critically polluted water bodies

Signs:

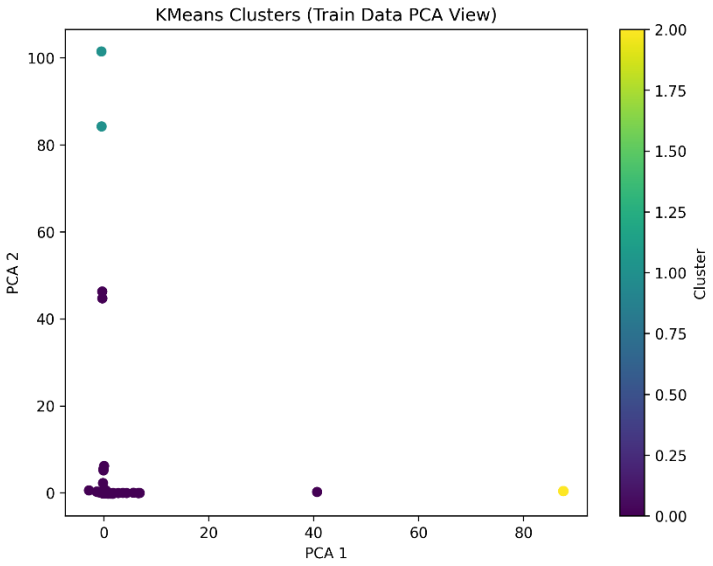
Very low DO: Aquatic life stress
Extremely high BOD: Massive organic waste load
Extremely high bacterial contamination: Raw sewage / untreated waste discharge

Likely reflects highly polluted locations near:

Untreated sewage outlets

Industrial discharge points

Stagnant wastewater zones



4.6.3 Water Quality Index (WQI) Classification

A Water Quality Index (WQI) was computed by assigning weighted contributions to key water quality parameters, including dissolved oxygen, biological oxygen demand, nitrate concentration, faecal coliform, and total coliform, to represent overall water quality status. Based on the calculated WQI values, water samples were classified into polluted and non-polluted categories using a threshold-based approach. These classified labels were then used to train a Random Forest Classifier for supervised learning analysis. This method enabled effective assessment of water quality by combining multiple environmental indicators into a single classification framework.

4.7 Correlation Analysis (Heat map)

The correlation analysis reveals that microbial contamination (faecal and total coliforms) is strongly interdependent, indicating widespread sewage intrusion across water bodies. Moderate correlation between nitrate and pH suggests agricultural runoff as a secondary pollution source. Weak correlations among most physicochemical parameters highlight the heterogeneous and multi-source nature of water pollution in Indian aquatic systems. The negative association between DO and BOD further confirms localized organic pollution and oxygen depletion

1. Strong positive correlations

Faecal coliform ↔ Total coliform (0.86)

- This is a **very strong positive correlation**

- Meaning: whenever faecal contamination increases, total bacterial contamination also increases

Conclusion:

- This clearly indicates **sewage contamination**
- Indian rivers/lakes with high values here are likely affected by:
    - untreated wastewater
    - poor sanitation infrastructure

pH ↔ Nitrate (0.61)

- Moderately strong positive correlation

Conclusion:

- Areas with higher nitrate also show changes in pH
- This suggests:
    - **agricultural runoff (fertilizers)**
    - possible **eutrophication risk**

2. Weak or Near-Zero Relationships

Most other correlations are close to **0**, meaning:

- Temperature vs pollution → very weak
- Nitrate vs coliform → almost independent
- BOD vs nitrate → weak

Conclusion:

- Pollution sources are **diverse and independent**
- No single factor explains overall pollution
- Indian water bodies are affected by **multiple localized sources**

3. Important Negative Relationship

DO ↔ BOD (-0.22)

- Slight negative correlation

Scientific meaning:

- When **BOD (Biochemical Oxygen Demand)** increases,
- **DO (Dissolved Oxygen)** decreases

Conclusion:

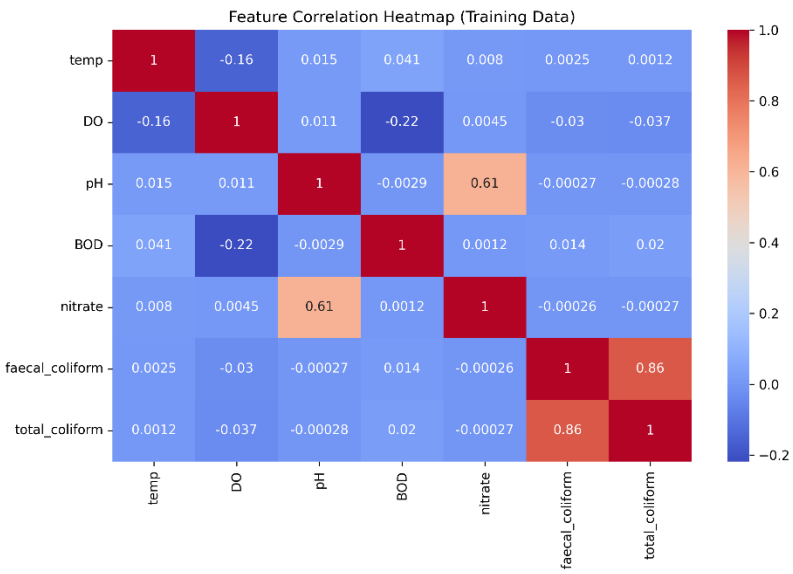
- This is a classic sign of **organic pollution**
- Indicates:
    - industrial discharge
    - sewage input

4. Temperature Effects

- Temperature has **very weak correlations with all features**

Conclusion:

- Pollution is **not strongly seasonal** in this dataset
- Instead, it's driven more by **human activities than climate**



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4.8 Anomaly Detection / Classification Models

4.8.1 Anomaly Detection

Isolation Forest anomaly detection identified approximately 10% of water-quality samples as anomalous, revealing the presence of rare and unusual water-quality conditions within the dataset. The majority of samples followed consistent pollution patterns, suggesting chronic contamination across Indian water bodies, while anomalous observations likely represent localized severe pollution events, hidden biological contamination, or data inconsistencies. These findings demonstrate that unsupervised anomaly detection can complement traditional threshold-based assessments by uncovering atypical pollution signatures that may otherwise remain undetected.

4.8.2 Classification Model (Polluted vs Safe Water)

The Random Forest classifier achieved high predictive performance (98.48% accuracy), indicating that water pollution status can be effectively modelled using physicochemical and biological parameters. However, the classification relies on rule-based labelling derived from threshold conditions, which introduces bias and potential data leakage. While the model successfully captures structured pollution patterns such as organic and microbial contamination, it lacks generalization to real-world variability and does not differentiate between types of water bodies. Future work should incorporate unsupervised anomaly detection, water body-specific modelling, and regulatory-standard-based labelling for improved robustness.

4.9 Relevance of Machine Learning Techniques to Physical Sciences

The machine learning techniques applied in this study—such as clustering, regression, and anomaly detection—are widely used in the analysis of complex physical systems. Although the present work focuses on environmental data, the underlying methodologies are domain-independent and applicable to other scientific fields.

In high-energy physics, particularly in the study of particle interactions, large multivariate datasets are analysed to identify patterns, classify events, and detect rare phenomena. Techniques such as unsupervised clustering are used for event categorization, while anomaly detection methods are employed to identify rare or statistically significant events.

The approach used in this study for detecting anomalous water quality samples is conceptually similar to identifying rare particle interactions in experimental physics. This highlights the broader applicability of data-driven methods in scientific research beyond environmental studies.

5. Results:

The analysis of water quality data across various aquatic systems in India reveals significant spatial variation in pollution levels, strongly influenced by anthropogenic activities and environmental factors.

Small water bodies such as canals, tanks, and drainage systems were found to be the most polluted. A large proportion of these samples exhibited **Biochemical Oxygen Demand (BOD) values exceeding 5 mg/L**, along with elevated nitrate concentrations and significant presence of faecal coliforms. These values exceed both drinking water and marine life safety thresholds, indicating severe organic and microbial contamination.

Rivers located in proximity to urban and industrial regions also showed considerable pollution. Many samples recorded **dissolved oxygen (DO) levels below the recommended threshold of 5 mg/L**, along with high BOD and coliform concentrations. However, some river samples demonstrated acceptable water quality, suggesting that relatively unpolluted segments still exist, particularly in regions with minimal human interference.

Groundwater analysis indicated moderate contamination in several regions, primarily associated with **elevated nitrate concentrations and conductivity levels**, likely resulting

from agricultural runoff and industrial discharge. While BOD levels in groundwater were generally lower compared to surface water bodies, the presence of chemical pollutants raises concerns regarding long-term suitability for drinking purposes.

The machine learning-based analysis further supported these findings. Clustering techniques identified distinct groups corresponding to **low, moderate, and high pollution levels**, with a significant number of samples falling into moderate-to-high pollution categories. Anomaly detection methods highlighted extreme cases of contamination, typically characterized by **very high BOD, low DO, and elevated coliform levels**, indicating critical ecological stress zones.

Overall, the results confirm that a substantial fraction of India's water bodies do not meet the standard criteria for safe drinking water or healthy aquatic ecosystems, emphasizing the urgent need for improved monitoring and pollution control measures.

6. Implications of theory:

Water quality in India remains a critical issue due to the rapid industrialization, urbanization, and agricultural practices. The state-wise and river-wise analysis highlights the diverse challenges faced by different regions. Addressing these issues requires a multifaceted approach involving government initiatives, industrial compliance, and public participation. Sustainable management and proactive measures are essential to ensure the availability of clean and safe water for future generations.

7. Limitations

This study provides a broad analysis of the condition of water bodies in India, including major rivers, small rivers, lakes, ponds, tanks, canals, groundwater, and water treatment plants. It offers useful insights based on important water quality parameters such as temperature, dissolved oxygen, Biological Oxygen Demand (BOD), nitrate concentration, faecal coliform, and total coliform collected over approximately 4–5 years (2012–2019). The analysis helps in identifying general pollution patterns and raises awareness about human activities contributing to water resource degradation, emphasizing the need for timely corrective measures.

However, the study has several limitations. The analysis represents a **Level 1 investigation**, which provides generalized pollution assessment rather than highly precise, location-specific conclusions. The Water Quality Index (WQI) used in this project is based on simplified weighting assumptions and may not fully capture the actual environmental complexity of different water bodies. The results primarily indicate whether water is polluted and suggest probable contributing factors rather than establishing direct causation.

The machine learning models used in this project are also subject to limitations. Their performance depends heavily on the quality, completeness, and representativeness of the dataset. Since the dataset covers limited parameters, some important pollution indicators such as heavy metals, industrial chemical discharge, seasonal variations, rainfall impact, and geographical influences were not included. This may reduce the predictive accuracy and generalizability of the models. The anomaly detection model (Isolation Forest) identifies unusual patterns statistically, but these anomalies may not always correspond to real-world pollution events. Similarly, clustering and classification results depend on feature scaling, parameter selection, and assumptions made during pre-processing.

Another limitation is that the dataset represents historical observations and may not fully reflect current water quality conditions. Water pollution is dynamic and influenced by rapidly changing environmental and anthropogenic factors.

For more actionable and precise insights, future studies should incorporate real-time monitoring data, additional water quality parameters, geographic information, and deeper category-specific analysis. Advanced hybrid machine learning approaches and domain-specific environmental validation could further improve prediction reliability and practical applicability.

7. References

2. *https://en.wikipedia.org/wiki/List_of_rivers_of_India
3. https://en.wikipedia.org/wiki/List_of_major_rivers_of_India
4. https://en.wikipedia.org/wiki/Water_pollution_in_India
5. https://en.wikipedia.org/wiki/Water_pollution
6. https://en.wikipedia.org/wiki/Water_resources_in_India
7. https://en.wikipedia.org/wiki/Pollution_of_the_Ganges
8. https://savetherivers.in/river-pollution-in-india/
9. Water pollution in rivers: A long pathway to a better tomorrow (https://www.downtoearth.org.in/pollution/water-pollution-in-rivers-a-long-pathway-to-a-better-tomorrow-95987)
10. Analysis of water quality of rivers by central pollution control board (https://pib.gov.in/PressReleseDetailm.aspx?PRID=1941065)
11. Current Status of Ponds in India: A Framework for Restoration, Policies and Circular Economy (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9638415/)

12. Water Pollution: How it affects Lakes & Ponds and Its Sources (https://www.solitudelakemanagement.com/water-pollution-how-it-affects-lakes-ponds-and-its-sources/)
13. What's causing mass fish death in India's ponds and lakes (https://india.mongabay.com/2022/07/explainer-whats-causing-mass-fish-deaths-in-indias-ponds-and-lakes/)

14. Water quality monitoring in India achievements and constraints (https://unstats.un.org/unsd/environment/envpdf/pap_wasess5a2india.pdf)

15. *Dying Traditional Water Bodies in India Struggling to Survive against Unplanned Development (https://www.scirp.org/journal/paperinformation?paperid=85526)
16. Water Quality of Medium & Minor Rivers-2019 (https://cpcb.nic.in/wqm/2019/Water_Quality_data_of_River_2019.pdf)
17. River Water Quality Monitoring (https://indiawris.gov.in/wiki/doku.php?id=river_water_quality_monitoring)
18. Status of water treatment plants in India (https://cpcb.nic.in/openpdffile.php?id=UmVwb3J0RmlsZXMvTmV3SXRlbV8xMDNfc3RhdHVzb2Z3YXRlcnF1YWxpdHlwYWNrYWdlLnBkZg==)
19. Waste water treatment for water security in India (http://irapindia.org/images/irap-Occasional-Paper/Wastewater-Water-Security-India.pdf)
20. Water contamination: Still a serious national challenge (https://timesofindia.indiatimes.com/blogs/voices/water-contamination-still-a-serious-national-challenge/)
21. Temperature and water (https://www.usgs.gov/special-topics/water-science-school/science/temperature-and-water)
22. Indicators: Dissolved Oxygen (https://www.epa.gov/national-aquatic-resource-surveys/indicators-dissolved-oxygen)
23. Dissolved Oxygen and Water (https://www.usgs.gov/special-topics/water-science-school/science/dissolved-oxygen-and-water)
24. How does pH affect water quality (https://atlas-scientific.com/blog/how-does-ph-affect-water-quality/)

25. pH of Water (https://www.fondriest.com/environmental-measurements/parameters/water-quality/ph/#:~:text=If%20the%20pH%20of%20water,levels%20outside%20of%20this%20range.)
26. pH and Water (https://www.usgs.gov/special-topics/water-science-school/science/ph-and-water)
27. How does Conductivity affect Water Quality? (https://atlas-scientific.com/blog/how-does-conductivity-affect-water-quality/)
28. Water quality indicator: Electrical conductivity (https://www.qmul.ac.uk/chesswatch/media/chesswatch/Electrical-conductivity-leaflet.pdf)
29. Biochemical oxygen demand (https://en.wikipedia.org/wiki/Biochemical_oxygen_demand)
30. Biochemical Oxygen Demand(BOD) and Water (https://www.usgs.gov/special-topics/water-science-school/science/biochemical-oxygen-demand-bod-and-water)
31. Nitrate and nitrite contamination in drinking water and cancer risk: A systematic review with meta-analysis (https://www.sciencedirect.com/science/article/pii/S0013935122003152)
32. Nitrate in Drinking Water (https://www.health.state.mn.us/communities/environment/water/contaminants/nitrate.html)
33. Drinking Water Nitrate and Human Health: An Updated Review, Mary H. Ward,[1,*] Rena R. Jones,[1] Jean D. Brender,[2] Theo M. de Kok,[3] Peter J. Weyer,[4] Bernard T. Nolan,[5] Cristina M. Villanueva,[6,7,8,9] and Simone G. van Breda[3] (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6068531/)
34. Microbial pollution of water with special reference to coliform bacteria and their nexus with environment, Sudip Some [1], Rittick Mondal [2], Debasis Mitra [3], Divya Jain [4 5], Devvret Verma [5], Samanwita Das (https://www.sciencedirect.com/science/article/pii/S2772427121000085)
35. Human Fecal Pollution Monitoring and Microbial Risk Assessment for Water Reuse Potential in a Coastal Industrial-Residential Mixed-Use Watershed, Akechai Kongprajug,[1,†] Thammanitchpol Denpetkul,[2,†] Natcha Chyerochana,[1] Skorn Mongkolsuk,[1,3] and Kwanrawee Sirikanchana, (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8093506/)

36. Fecal Coliform bacteria in Water, (https://www.knowyourh2o.com/outdoor-4/fecal-coliform-bacteria-in-water)
37. Fecal Pollution of Water, (https://cfpub.epa.gov/si/si_public_record_report.cfm?Lab=NERL&dirEntryId=196784)
38. Fecal Contamination and High Nutrient Levels Pollute the Watersheds of Wujiang, China, by **Raju Sekar, Xin Jin, Shuang Liu, Jing Lu, Jianwei Shen, Yingya Zhou, Yingya Zhou, Ziyang Gong, Xueying Feng, Shengjie Guo and Wenlong Li,** (**https://www.mdpi.com/2073-4441/13/4/457**)